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Outdoor mean radiant temperature estimation in the tropical urban environment

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ABSTRACT

A large scale estimation of mean radiant temperature (t_{mrt}) is conducted at two sites using customised globe thermometers. The measurement points cover a variety of urban typologies such as high-rise offices, parks, large water bodies and housing apartments. Data is derived using a t_{mrt} formula calibrated to the local climate. Measurements for clear, sunny days are used for the analysis of the average diurnal t_{mrt} profile.

The diurnal t_{mrt} profile shows that the t_{mrt} differential between points is most evident during daytime, and is affected most significantly by shade cast by trees and buildings. Results also show that common urban constituents such as greenery and large water bodies, while proven to effectively reduce the ambient temperature of its surroundings throughout the day, do not affect t_{mrt} significantly after nightfall. Further analysis reveals a correlation between sky view factor and t_{mrt} in the day. Measurement points in different parks exhibit contrasting trends in t_{mrt} reduction.

Results of the study also provide a realistic threshold for the lowering of outdoor t_{mrt}. Trees, shrubs and green walls may be introduced into the outdoor environment with the intention of reducing t_{mrt} to a desirable level for a specific time range.

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1. Introduction

Modern civilisation has improved our lives in many ways. It has also produced a new environment, creating issues of adaptation. These issues include global warming, industrial waste, and pollution. More people are vulnerable to urbanisation problems as the ever increasing urban population, which was estimated at 48% or three billion, is expected to be five billion by 2030 [1]. The projected global average surface warming at the end of the 21st century is between the range of 0.3 °C–6.5 °C [2]. The rise in temperature will have a direct impact on the quality of outdoor spaces in urbanised areas.

Outdoor spaces are important as it encompasses pedestrian traffic as well as various outdoor activities. Increased outdoor activity in urbanised areas can generate many positive attributes [3,4]. Therefore, it is important for outdoor spaces to be properly designed. The outdoor microclimate is an important factor that determines the quality of outdoor urban spaces as it affects thermal comfort and subsequent usage [5].

There are several methods of determining the quality of both the indoor [6,7] and outdoor [8] microclimate. The use of biometeorological indices has enabled quantification of thermal comfort and assessment in tandem with behavioural aspects. Useful heat stress indices have also been developed to describe thermal stress [9–11]. According to the rational approach [10], the evaluation of thermal environments by means of a suitable comfort [6] or stress index [9,10] requires the measurement of four physical quantities of the air temperature, the mean radiant temperature, the air velocity and the relative humidity. Among them, one of the main factors contributing to the thermal response of man to his surrounding environment is the mean radiant temperature (t_{mrt}). This quantity plays a crucial role not only in indoor situations but also outdoors as indicated in several studies which have stressed that outdoor thermal comfort is highly dependent on the short wave and long wave radiation fluxes from the surroundings [12,13].The estimation of t_{mrt} can be done by two-sphere radiometers, globe thermometers, constant-air-temperature sensors [14]. Calculation of t_{mrt} is also possible using radiant fluxes and the angle factors of surrounding surfaces [14,15].

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The effects of t_{mrt} can be studied through numerical modelling [16,17]. This method is useful when used for iterative studies such as the comparison of width-to-height ratios and orientations of street canyons. However, model geometry and ambient conditions are often simplified. Recent developments in solar and long wave environmental irradiance modelling takes into account large-scale urban geometry as well as important urban components such as tree and shrubs [18].

The estimation of radiant temperature is often an integral component to the assessment of thermal environments. Common thermal assessment indices such as the Physiological Equivalent Temperature (PET) and the recently developed Universal Thermal Climate Index (UTCI) are evaluated with t_{mrt} as a variable component [19–21]. Other heat stress indices such as the Wet Bulb Globe Temperature (WBGT) consider radiant temperature in the form of the globe surface temperature [22,23]. The fluctuating radiation fluxes due to complex environments have contributed to the uncertainty in t_{mrt} estimation, and studies have shown that it can affect the overall thermal assessment in some temperature ranges [24]. Therefore, it is important that the estimated t_{mrt} can accurately reflect the prevalent conditions of radiation fluxes of the measured space.

The urban environment can be described as an amalgamation of buildings, vegetation, water bodies and many other constituents. It is therefore important for us to understand how the behaviour of t_{mrt} is influenced by these factors in the outdoor urban environment.

The purpose of this study is to observe the diurnal t_{mrt} profile of the outdoor environment and to identify any relation between t_{mrt} and the corresponding urban typology.

2. Methodology

2.1. Calibration of customised globe thermometers

Customised globe thermometers are used to estimate the t_{mrt} for all measurement points. In order to ensure the accuracy of t_{mrt} estimation, readings from the customised globe thermometers are first estimated against readings from a net radiometer. The mean convection coefficient of the formula used for t_{mrt} estimation is recalibrated for use in the local context.

The mean radiant temperature is defined as the 'uniform temperature of an imaginary enclosure in which the radiant heat transfer from the human body equals the radiant heat transfer in the actual non-uniform enclosure' [25]. It is one of the meteorological parameters governing human energy balance and human thermal comfort.

The measurement of t_{mrt} is done via the use of a globe thermometer [26–29]. The globe thermometer was first developed for indoor measurements, but has later been applied outdoors [30]. The Vernon globe is a 150 mm diameter copper sphere painted black with a thermometer positioned in the middle of the sphere. For convenience, smaller globes were developed. The 38 mm globe thermometer is a common option as the globe used is a table tennis ball, which can be readily purchased and conveniently replaced [31]. The accuracy of the 38 mm globe thermometer can be adjusted to cater to outdoor conditions by recalibrating the mean convection coefficient. This method has been tested in Sweden [32] and shown to be effective in outdoor conditions.

To ensure validity of the t_{mrt} estimation for this study, the mean convection coefficient of the formula for t_{mrt} estimation is recalibrated. This is done by comparing the estimates from the customised globe thermometer to the long wave and short wave readings from a net radiometer.

Two sets of measurements are made, one at each of the study areas (Table 1).

Two different methods for measuring the t_{mrt} outdoors are put to comparison:

1. Method A – Radiant flux measurements, where t_{mrt} calculation is based on short wave and long wave angular factors for a sphere;
2. Method B – 40 mm flat grey globe thermometer with t_{mrt} equation from ISO 7726:1998 [14].

Results are used to recalibrate the t_{mrt} formula for Method B, so that the mean convection coefficient in the t_{mrt} equation will be representative of local outdoor conditions. Recalibration is done via statistical analysis using the IBM SPSS software [33].

Measurements are then made in another area with Methods A and B. The t_{mrt} formula used for the second measurement will be with the recalibrated mean convection coefficient. This is done to ensure validity of the recalibration.

Measurements are taken at the frequency of one minute, and averaged to five-minute intervals [32]. Data gathered from Methods A and B will be used to recalibrate the globe thermometer to improve the accuracy of the globe thermometer with respect to radiant flux measurements. Table 2 shows the measured variables and equipment used.

The instrument setup used for the radiant flux measurements is shown in Fig. 1. A net radiometer with three integrated pyranometer and pyrgeometer arms (Kipp and Zonen, CNR 4), each measuring both incoming and outgoing short wave and long wave fluxes, are mounted on a steel tripod stand to measure the three-dimensional radiation field. Short wave and long wave radiation fluxes from the four cardinal points (North, East, South and West), as well as those from the upper and lower hemisphere, are measured. The newly purchased net radiometer was factory calibrated. The pyranometers were calibrated side by side to a reference CMP 3 pyranometer according to ISO 9847:1992 annex A.3.1. The pyrgeometers were calibrated side by side to a reference CG(R) 3 pyrgeometer [34].

In a previous study conducted in Sweden [32], the globe thermometer used consisted of a hollow acrylic sphere coated in flat grey paint (RAL 7001), with a diameter of 38 mm and a thickness of 1 mm, with a Pt100 sensor at its centre [28,29,31]. The 38 mm flat grey globe thermometer was mounted on the micrometeorological station.

Table 1
Measurement period.

	Study Area 1 Green Technology Laboratory rooftop	Study Area 2 School of Design and Environment Block 1 rooftop
Measurement date	28th February 2011 to 18th March 2011 30th March 2011 to 12th April 2011	13th August 2011 to 13th September 2011
No. of days measured	33	32
Purpose of measurement	Recalibration of mean convection coefficient to fit study	Validation of t_{mrt} calculation using recalibrated mean convection coefficient
No. of days used for recalibration of mean convection coefficient	33	–
No. of days used for validation of mean convection coefficient	–	32

Table 2
Measured variables and equipment.

Variable	Equipment	Measurement range	Accuracy
Air temperature, t_a	*HOBO U12-012 Temp/RH data logger*	−20 °C to 70 °C	±0.35 °C from 0 °C–50 °C, to a maximum of ±3.5%
Globe temperature, t_g	*HOBO Thermocouple Data Logger*, U12-014 with Type-T Copper-Constantan thermocouple sensors and 40 mm diameter ping pong ball	−200 °C to 100 °C	±1.5 °C
Wind speed, V_a	*Onset Wind Speed Smart Sensor*, S-WSA-M003	0 m s^{-1}–45 m s^{-1}	±1.1 m s^{-1} or ±4% of reading, whichever is greater
Short and long wave radiation, K, L	*Kipp & Zonen*, CNR 4 with integrated pyranometer, pyrgeometer, Pt-100 and thermistor	Pyranometer: 0 W m^{-2}–2000 W m^{-2} Pyrgeometer: −250 W m^{-2} to 250 W m^{-2} Pt-100: −200 °C to 600 °C Thermistor: −40 °C to 300 °C	Pyranometer: <5% uncertainty (95% confidence level) Pyrgeometer: <10% uncertainty (95% confidence level) Pt-100/Thermistor: ±0.7 °C

In this study, the globe thermometer is made of a 40 mm ping pong ball with a HOBO thermocouple wire at its centre. The ping pong is coated in flat grey paint (Phylox Nippon 144 dove grey). The 40 mm ping pong ball is preferred over the original 38 mm due to the decrease in availability of 38 mm ping pong balls [35]. The 40 mm flat grey globe thermometer is secured on the arm of one of the net radiometer sensors to ensure accuracy of readings for both methods of t_{mrt} calculation. A total of nine other customised globe thermometers are set up near the globe thermometer to ensure measurement consistency. All thermocouple loggers and Type-T Copper-Constantan thermocouple sensors used for this study are new and factory calibrated. The globe thermometers are positioned such that they will not experience any effects of overshadowing from the mounting structure.

2.2. Characteristics of Site A and Site B

An urban scale mapping of t_{mrt} is conducted after the recalibration of the t_{mrt} formula with the intention of understanding the diurnal t_{mrt} profile of highly urbanised areas. The measurement areas are categorised into the following:

- Area with high density commercial buildings;
- Area in close proximity to a large water body;
- Area with high density residential buildings;
- Park.

Singapore has a tropical rainforest climate with no distinctive seasons. Near-surface air temperature usually ranges from 23 °C to 32 °C. April and May are the hottest months, and the monsoon season from November to March [36]. The first site, Site A, is located in the Central Business District (CBD) of Singapore. The second site, Site B, is in the residential area of eastern Singapore. The locations of the measurement points are shown in Fig. 2.

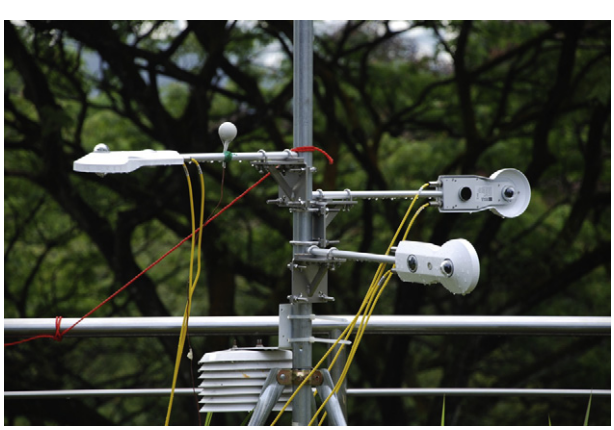

Fig. 1. 40 mm globe thermometer mounted on net radiometer.

The characteristics of both sites are shown in Table 3.

Measurements are conducted for a total of four months in 2012. Only days of clear, sunny weather are used for analysis (Table 4). The term clear, sunny weather is defined by scrutinising the hourly solar irradiance profile and ensuring a smooth curve of at peak of at least 700 W m^{-2} at the hottest time of the day. Measurements are made for the variables shown in Table 5.

3. Results and discussion

3.1. Validation of customised globe thermometers

Calculation of t_{mrt} can be done for Method A using the following method. The mean radiant flux density (S_{str}) of the human body is first determined. In order to calculate S_{str}, the six individual measurements of the short wave radiation and long wave radiation fluxes have to be multiplied by the angular factors F_i (i = 1–6) between a person and the surrounding surfaces according to Equation (1) [37]:

$$S_{str} = \alpha_k \sum_{i=1}^{6} K_i F_i + \varepsilon_p \sum_{i=1}^{6} L_i F_i \tag{1}$$

K_i = Short wave radiation fluxes (i = 1–6) (W m^{-2})
L_i = Long wave radiation fluxes (i = 1–6) (W m^{-2})
F_i = Angular factors between a person and the surrounding surfaces (i = 1–6)
α_k = Absorption coefficient for short wave radiation (standard value 0.7)
ε_p = Emissivity of the human body. According to Kirchhoff's laws ε_p is equal to the absorption coefficient for long wave radiation (standard value 0.97)

F_i is dependent on the position and orientation of the observer [6]. The calculation of F_i is complicated for complex urban forms and simplifications are thus necessary. For a sphere, F_i is 0.167 for all six directions.

The t_{mrt} (°C) can be calculated from the Stefan–Boltzmann equation:

$$t_{mrt} = \sqrt[4]{\left(\frac{S_{str}}{\varepsilon_p \sigma}\right)} - 273.15 \tag{2}$$

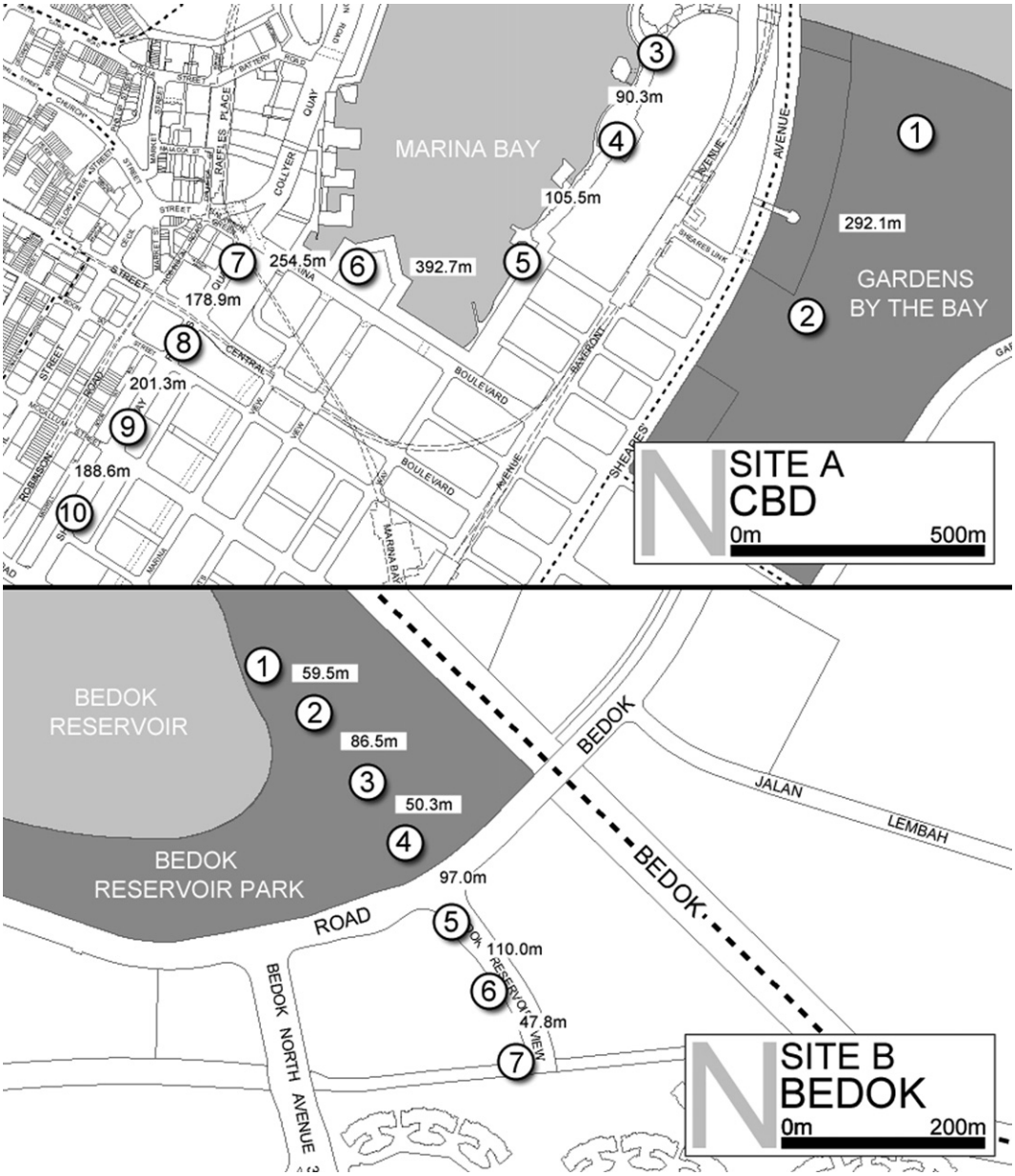

Fig. 2. Sites A and B.

Table 3
Site characteristics.

Site	A			B	
Location	Shenton way	Marina Bay Sands Casino	Gardens by the bay	Bedok reservoir park	Bedok reservoir view
Characteristic	High density commercial buildings	Large water body	Park	Park	High density residential
Number of measurement points	4	4	2	4	3

Table 4
Measurement period.

	Site A	Site B
Measurement period	March–April 2012	April–June 2012
Dates used for analysis	15/03/2012	13/04/2012
	16/03/2012	16/04/2012
	17/03/2012	23/04/2012
	01/04/2012	26/04/2012
	04/04/2012	27/04/2012
	13/04/2012	28/04/2012
	15/04/2012	09/05/2012
	18/04/2012	10/05/2012
	23/04/2012	16/05/2012
	–	06/06/2012

Table 5
Equipment list.

Variable	Instrument	Logging interval	Averaged to
Air temperature, t_a	*HOBO Thermocouple Data Logger*, U12-014	1 min	5 min
Globe temperature, t_g	*HOBO Thermocouple Data Logger*, U12-014	1 min	5 min
Wind speed, V_a	*HOBO Wind Speed Smart Sensor*, S-WSA-M003	1 min	5 min
Sky view factor, SVF	Nikon *D80 Digital SLR camera*	–	–
Surface temperature, t_s	*Fluke Thermal Imager*	–	–

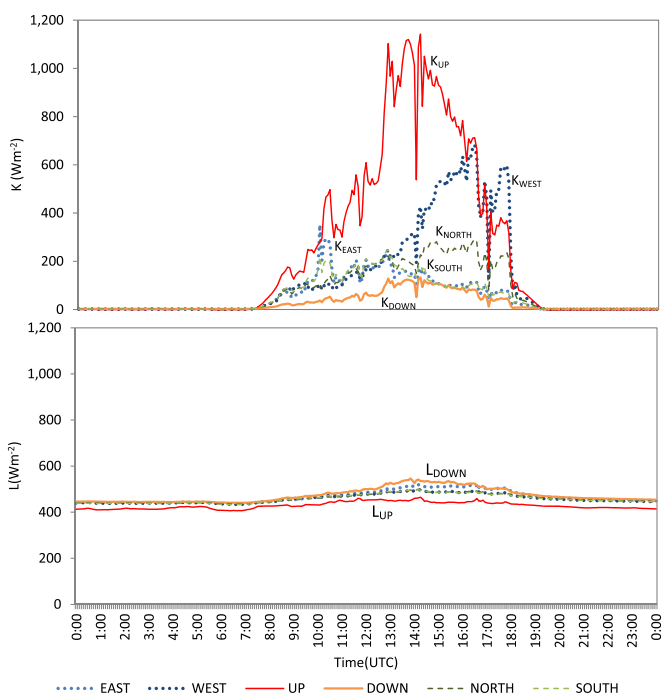

Fig. 3. Diurnal short wave (K) and long wave (L) profile for 18th March 2011.

S_{str} = Mean radiant flux density (W m^{-2})
σ = Stefan–Boltzmann constant ($5.67 \cdot 10^{-8}$ W m^{-2} K^{-4})
ε_p = Emissivity of the human body. According to Kirchhoff's laws εp is equal to the absorption coefficient for long-wave radiation (standard value 0.97)

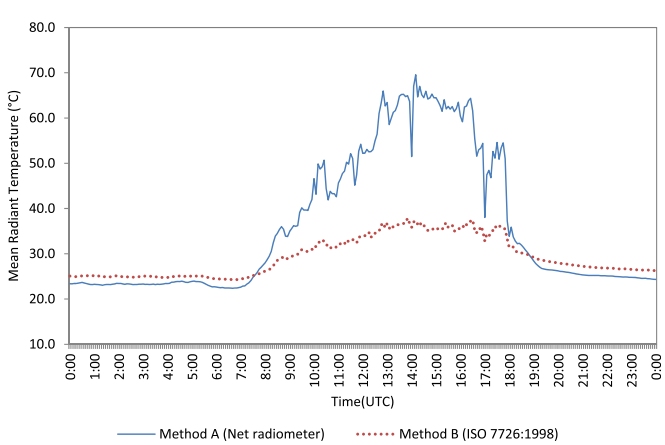

Fig. 4. Calculation of t_{mrt} using Method A and B (ISO 7726:1998) – 18th March 2011.

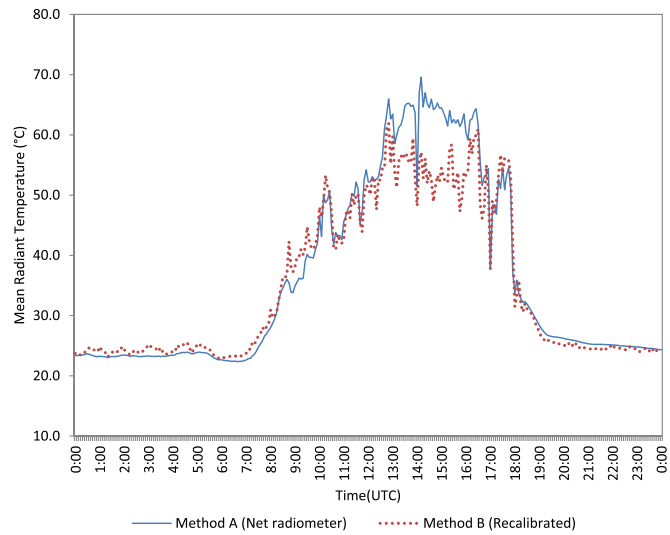

Fig. 5. Calculation of t_{mrt} using Method A and B (recalibrated) – 18th March 2011.

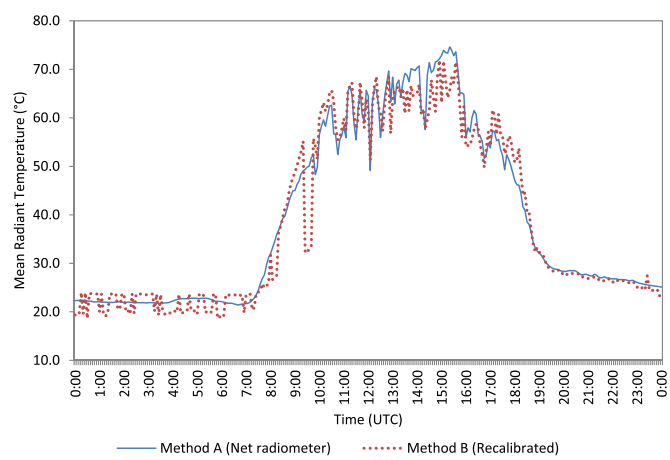

Fig. 6. Calculation of t_{mrt} using Method A and B (recalibrated) – Study Area 2, 16th August 2011.

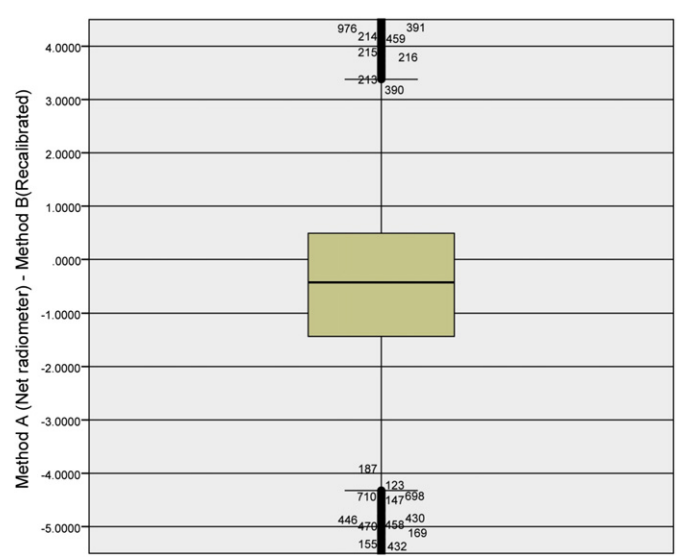

Fig. 7. Whisker plot for Study Area 2 (Method A – Method B).

Estimation of t_{mrt} can be done for Method B using the following method. The globe temperature represents the weighted average of radiant and ambient temperatures. If the globe temperature, air temperature and air velocity are known then the t_{mrt} can be calculated according to Equation (3) [14]:

$$t_{mrt} = \left[(t_g + 273.15)^4 + \frac{1.1 \times 10^8 V_a^{0.6}}{\varepsilon D^{0.4}} \times (t_g - t_a) \right]^{0.25} - 273.15 \quad (3)$$

t_g = Globe temperature (°C)
V_a = Air velocity (m s^{-1})
t_a = Air temperature (°C)
D = Globe diameter (mm)
ε = Globe emissivity

The globe's mean convection coefficient $(1.1 \times 10^8 V_a^{0.6})$ is defined as the empirically derived parameter (1.1×10^8) and the wind exponent $(V_a^{0.6})$. Values of t_{mrt} for Method B are calculated using Equation (3).

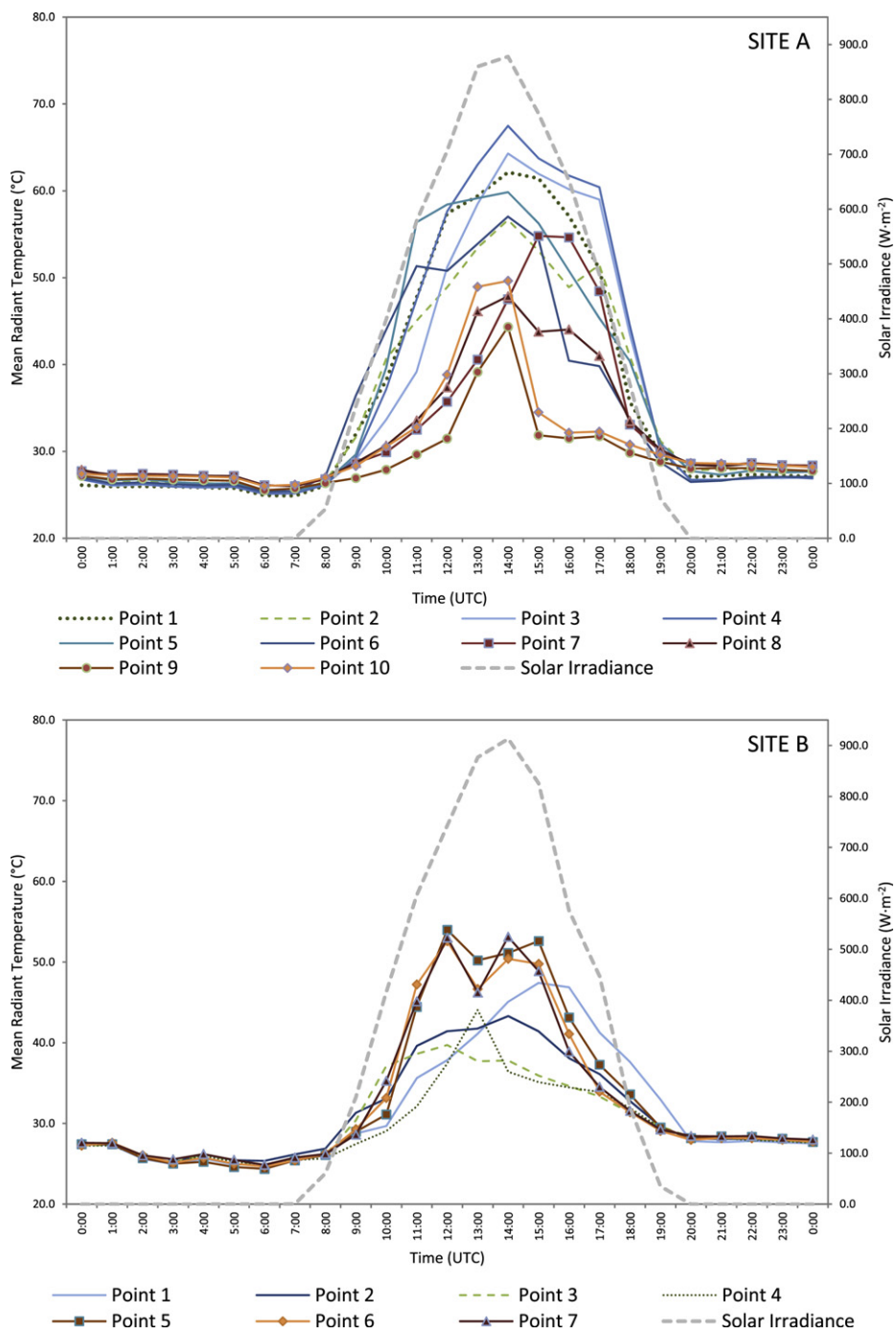

Fig. 8. Diurnal profiles of t_{mrt} for Site A and Site B.

A total of 65 days of field data from the two study areas are measured (Table 1). A typical clear sunny day is used for analysis of the diurnal t_{mrt} profile at Study Area 1. The 18th of March 2011 was a sunny day with clear skies. The mean near-surface air temperature was 27.3 °C, the minimum near-surface air temperature recorded was 24.2 °C and the maximum near-surface air temperature recorded was 31.0 °C. The mean wind speed was 3 m s^{-1}. The sun rose at 07:10 UTC and set at 19:16 UTC. The sun reached its maximum altitude of 87.5° at 13:13 UTC.

The diurnal short wave and long wave radiation profile is plotted for 18th March 2011. Fig. 3 shows the short wave and long wave radiation fluxes for six directions. K_{NORTH}, K_{SOUTH}, K_{EAST} and K_{WEST} represents the short wave radiation fluxes from the North, South, East and West respectively. K_{UP} represents short wave radiation from the upper hemisphere (facing the sky), and K_{DOWN} represents short wave radiation from the lower hemisphere (facing the ground).

The direct solar radiation K_{UP} reached its maximum of 1142 W m^{-2} at 14:10 UTC. Its value is most significantly affected by the position of the sun. The reflected short wave radiation K_{DOWN} follows a similar pattern to that of K_{UP} but with significantly lower values. It reached its maximum of 133 W m^{-2} at 14:10 UTC. K_{EAST} reached its maximum value of 351 W m^{-2} in the morning (10:00 UTC). Thereafter, readings remained low. In contrast, K_{WEST} remained low in the morning and increased steadily until it reached its maximum value of 684 W m^{-2} (almost twice the peak value of K_{EAST}) at 16:25 UTC. This is due to exposure to more intense direct solar radiation from the West. K_{NORTH} and K_{SOUTH} remained low throughout the day. K_{NORTH} reached its maximum value of 291 W m^{-2} at 16:25 UTC. K_{SOUTH} reached its maximum value of 246 W m^{-2} at 12:50 UTC. There is negligible short wave radiation activity in the absence of sunlight.

L_{NORTH}, L_{SOUTH}, L_{EAST} and L_{WEST} represents the long wave radiation fluxes from North, South, East and West respectively. L_{UP} represents long wave radiation from the upper hemisphere (facing the sky), and L_{DOWN} represents long wave radiation from the lower hemisphere (facing the ground).

The long wave fluctuations for all six directions follow a similar pattern. The L_{UP} profile is notably lower than the other profiles. This is due to the fact that L_{UP} is facing the sky and not directly at any objects. L_{UP} reached its maximum value of 465 W m^{-2} at 14:15 UTC. L_{DOWN} exhibited the highest profile, due to the fact that it is facing downwards and has maximum exposure to objects and surfaces. The difference between the other points is most evident during periods of intense exposure to sunlight. L_{DOWN} reached its maximum value of 546 W m^{-2} at 13:50 UTC. L_{EAST} increased steadily until 14:00 UTC. L_{WEST} does not exhibit a similar trend. This may be due to the fact that there is a wall on the West side of the net radiometer, and it is exposed to direct sunlight after 14:00 UTC. L_{EAST} reached its maximum value of 519 W m^{-2} at 16:30 UTC and L_{WEST} reached its maximum value of 498 W m^{-2} at 14:10 UTC. L_{WEST}, L_{NORTH} and L_{SOUTH} showed similar trends in fluctuation throughout the day. L_{NORTH} reached its maximum value of 499 W m^{-2} at 14:10 UTC and L_{SOUTH} reached its maximum value of 497 W m^{-2} at 13:50 UTC. The minimum value for all long wave radiation ranges from 433 W m^{-2} to 440 W m^{-2} at 06:40 UTC. There are slight fluctuations in the absence of sunlight.

Measurements using Method A (Net radiometer) are used to plot the diurnal t_{mrt} profile for 18th March 2011. The plot is overlaid with a plot of diurnal t_{mrt} using Method B. The t_{mrt} profile generated is similar to that of K_{UP}, indicating the high relevance of direct solar radiation to t_{mrt}. Fig. 4 shows that the calculation of t_{mrt} using the globe thermometer (Method B) differs from the values obtained via the net radiometer (Method A) drastically.

Results show that the measurement of t_{mrt} using the customised globe thermometers and Equation (3) is highly unsuited for use in the given context. There is a slight overestimate of t_{mrt} in the absence of sunlight and a drastic underestimate during sunlit hours. Any measurement of outdoor t_{mrt} using the customised globe thermometers using Equation (3) would be highly inaccurate.

Fig. 4 shows that the calculation of t_{mrt} using the globe thermometer (Method B) differs from the values obtained via the net radiometer (Method A) drastically. This is due to the mean convection coefficient used for the t_{mrt} formula (Equation (3)) that does not adequately represent the convective conditions found in the tropical outdoor climate.

Table 6
SVF values for measurement points in Site A and Site B.

Measurement point	Site A	Site B
1	0.646	0.270
2	0.711	0.341
3	0.668	0.614
4	0.671	0.178
5	0.602	0.486
6	0.803	0.664
7	0.340	0.668
8	0.512	–
9	0.240	–
10	0.275	–

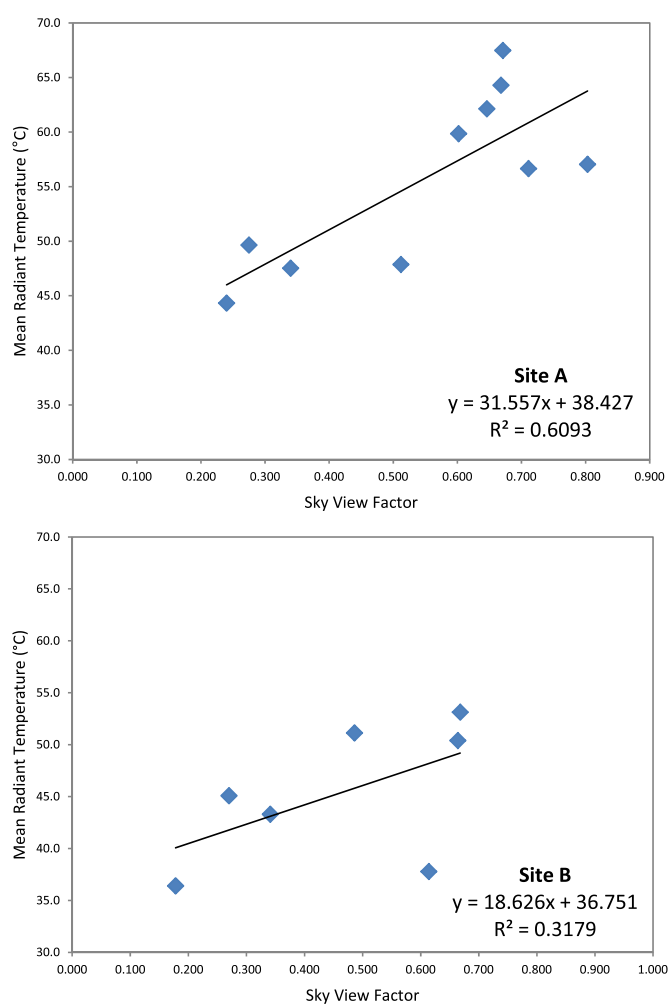

Fig. 9. Scatterplots of t_{mrt} and SVF for Site A and Site B at 14:00 UTC.

In order to utilise the formula in this context, the mean convection coefficient of $1.1 \times 10^8 V_a^{0.6}$ is recalibrated. This is done by considering the field measurements of the following variables:

- Mean radiant temperature (°C)
- Globe temperature (°C)
- Air temperature (°C)
- Air velocity (m s^{-1})

Only measurements taken from Study Area 1 are used for the calibration. A total of 33 days, which consists of diverse weather conditions, are used. The best fit curve generated gives the new mean convection coefficient of $3.42 \times 10^9 V_a^{0.119}$, giving us the recalibrated equation:

$$t_{mrt} = \left[(t_g + 273.15)^4 + \frac{3.42 \times 10^9 V_a^{0.119}}{\varepsilon D^{0.4}} \times (t_g - t_a) \right]^{0.25} - 273.15 \quad (4)$$

t_g = Globe temperature (°C)
V_a = Air velocity (ms^{-1})
t_a = Air temperature (°C)
D = Globe diameter (mm)
ε = Globe emissivity

The derived values are significant at 95% confidence interval.

A re-plot of the t_{mrt} profile for 18th March 2011 with Equation (4) shows the new t_{mrt} profiles. The new profile is much closer to

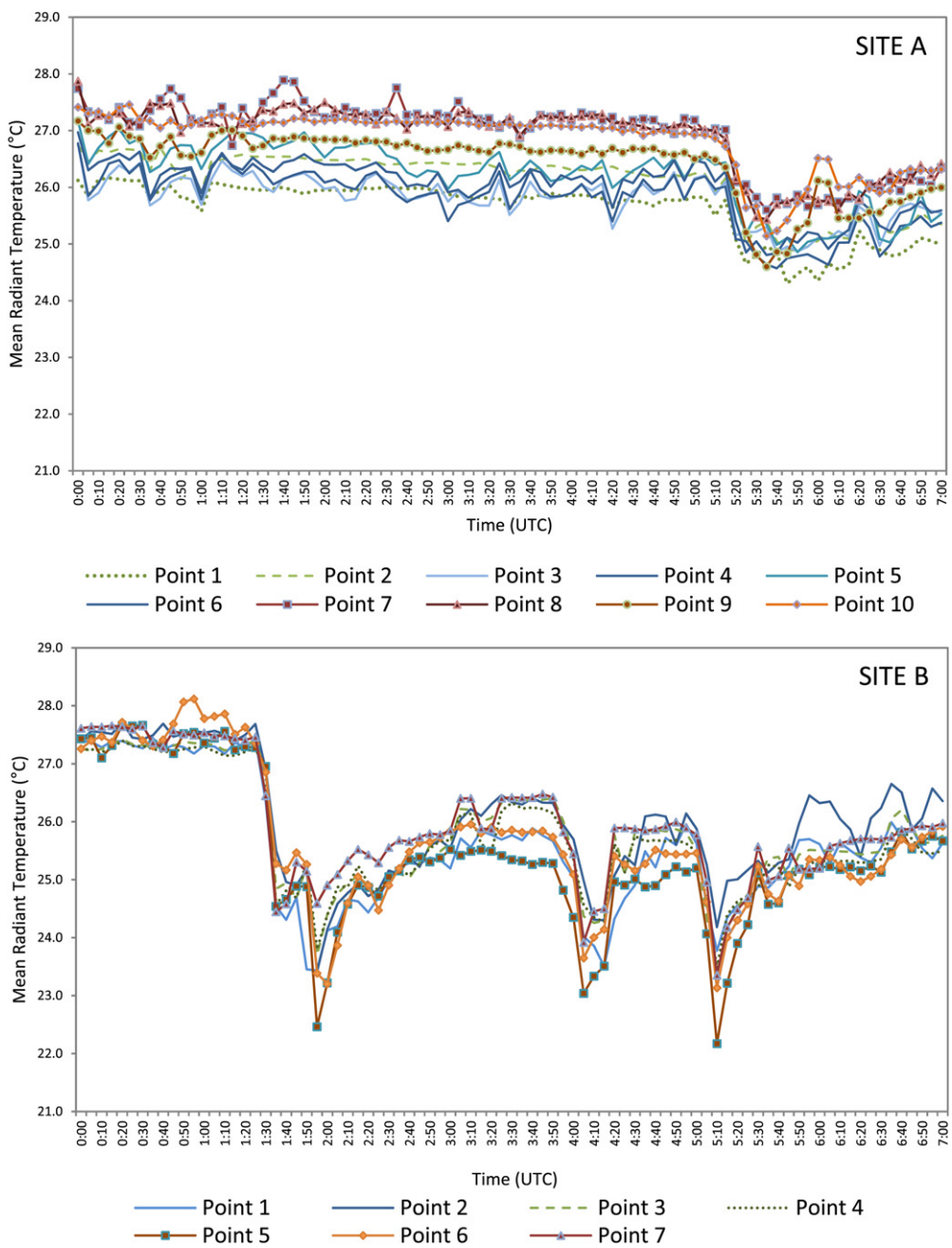

Fig. 10. Profiles of *tmrt* for Site A and Site B from 0:00 UTC to 07:00 UTC.

that of the t_{mrt} profile generated by Method A (Fig. 5). There is still a slight underestimation for certain periods during the day. The slight overestimation of t_{mrt} during night time is reduced to match the t_{mrt} of Method A. The profiles of the two methods show that the 40 mm globe thermometer has a response time of less than 5 min, which is similar to the 38 mm globe thermometers used in previous indoor tests [19]. There is a general trend of underestimation of t_{mrt} with increasing short-wave radiation. This suggests that the albedo of the globe may be slightly higher than desired, and may be reduced further by using a darker shade of grey.

To validate the new mean convection coefficient, Equation (4) is used to calculate the t_{mrt} of the same 40 mm globe thermometer at Study Area 2. A typical clear sunny day is chosen for analysis. A plot of the t_{mrt} profile for 16th August 2011 with Equation (4) shows the new t_{mrt} profile. Similar to the previous re-plot, the new profile is much closer to that of the t_{mrt} profile generated by Method A (Fig. 6).

Fig. 7 shows the whisker plot for the difference between t_{mrt} values derived from Method A and Method B. The median is approximately −0.4 °C, the inter-quartile range is approximately between 0.5 °C and −1.4 °C and the extreme values lie between 3.4 °C and −4.3 °C. This shows that most of the t_{mrt} values derived from Method B do not differ from that of Method A by 4.0 °C.

The 40 mm globe, together with the HOBO thermocouple data logger, can provide a good diurnal t_{mrt} profile that can be comparable to that of one derived through radiant flux measurements via a net radiometer (Figs. 5 and 6). The size of the globe does not demonstrate any significant impediment to its response time.

3.2. Diurnal t_{mrt} profiles of Site A and Site B

The days selected for analysis for both sites are shown in Table 3. The solar irradiance is plotted for all relevant days and the diurnal profile is used to determine the clear, sunny days to be used for analysis.

The diurnal t_{mrt} profiles of all ten-measurement points in Site A are shown in Fig. 8. Measurements are taken every minute and averaged to one hour intervals. Two distinctive trends can be observed. Points 1 to 6 exhibit significantly higher values of t_{mrt} in the day, especially during the periods of 08:00 UTC to 12:00 UTC and 14:00 UTC to 18:00 UTC. In comparison, Points 7 to 10 recorded lower values of t_{mrt}. The higher values recorded for Points 7 to 10 occur during the period of 11:30 UTC to 14:00 UTC. Points 1 to 6 are situated in the park and near a large water body, whereas Points 7 to 10 are located along a street surrounded by high rise commercial buildings.

The diurnal t_{mrt} profiles of all seven measurement points in Site B are also shown in Fig. 8. Measurements are taken every minute and averaged to one hour intervals. Three distinctive trends can be observed. In general, Points 3 and 4 exhibit the lowest t_{mrt} profiles during the day. Points 1 and 2 exhibit slightly higher readings, especially after 12:00 UTC. Points 5 to 7 exhibit the highest readings. Points 1 to 4 are situated in the park, whereas Points 5 to 7 are located along high rise residential buildings.

The measurements are plotted against solar irradiance. The difference in t_{mrt} of the maximum and minimum points for Sites A and B are 23.2 °C and 14.3 °C respectively.

3.3. Comparison with Sky View Factor (SVF)

The t_{mrt} profile for both sites are analysed with respect to the urban morphology. There is a significant difference in t_{mrt} profile between measurement points along the streets of high rise buildings and the other areas. The Sky View Factor (SVF), which is an indirect representation of built morphology, is used to identify any significant correlation. The SVF value for each measurement point is calculated by first taking a photograph of each point using a fisheye lens. Each photograph is subsequently processed using the Rayman software [38].

The SVF values for all measurement points in Site A and Site B are shown in Table 6. A scatterplot is made for the hottest time of the day for both sites. Scatterplots of the SVF and t_{mrt} values show that there is a correlation between SVF and t_{mrt} (Fig. 9).

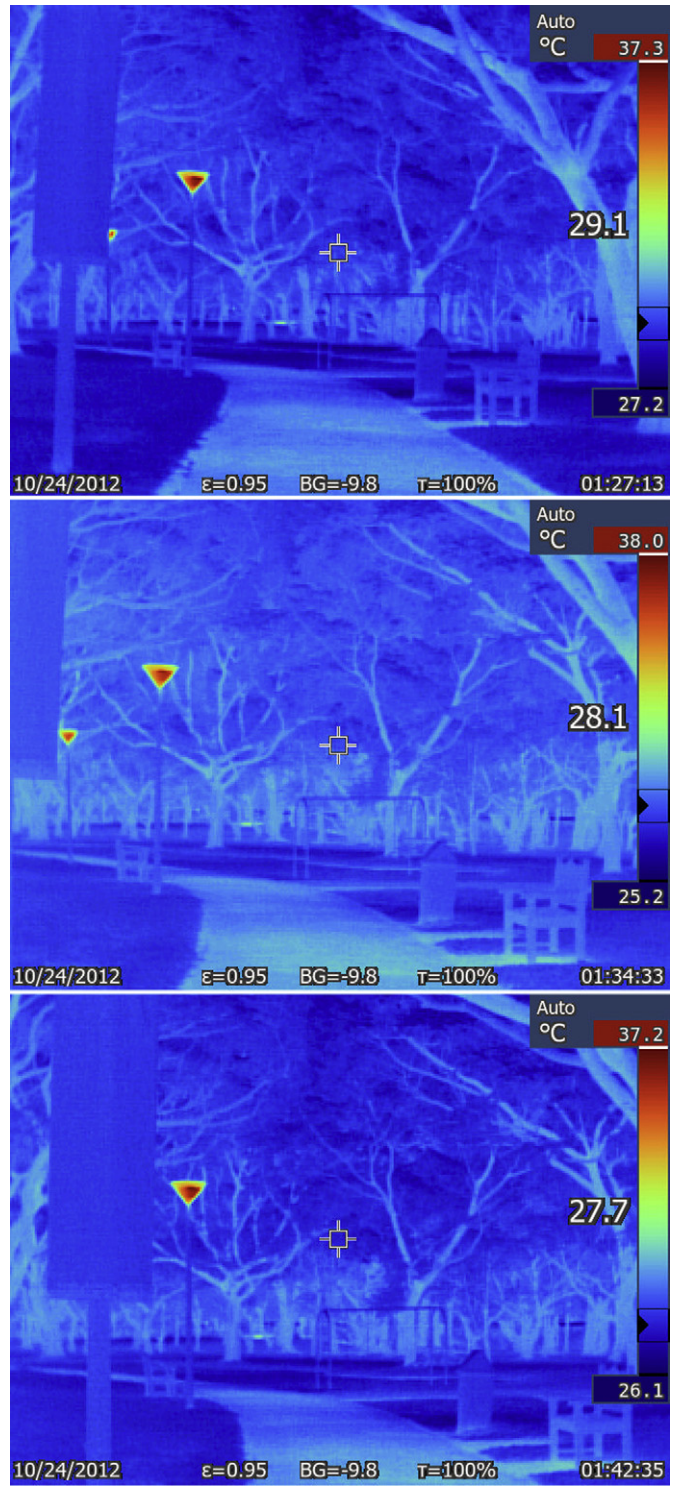

Fig. 11. Thermal imaging of tree surface temperature for Site B.

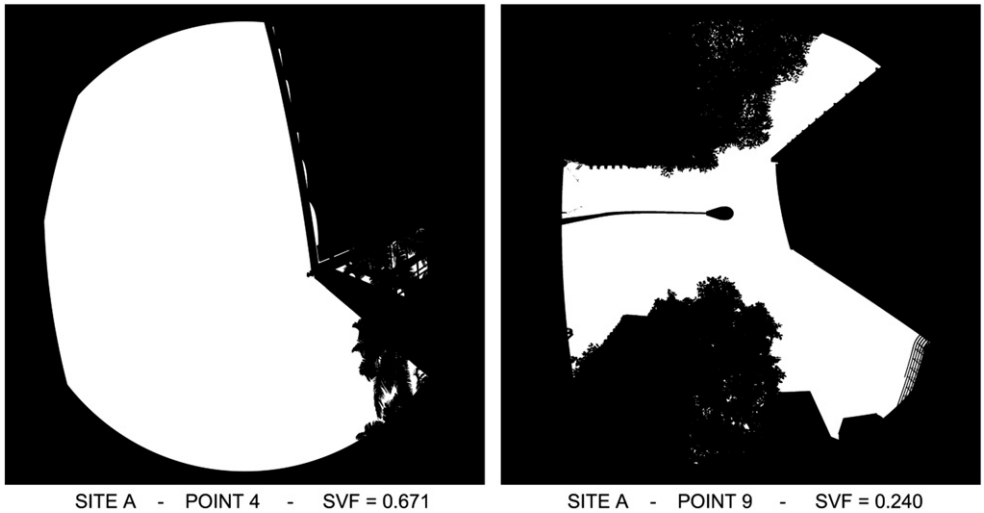

Fig. 12. SVF of Point 4 and Point 9.

3.4. Influence of urban constituents on t_{mrt}

Fig. 8 shows that buildings affect the t_{mrt} differently for both sites. While the t_{mrt} is the lowest along the high-rise buildings for Site A, the reverse is true for Site B. Although the measurement points are surrounding by high-rise buildings, there is a significant difference between the SVF values for the points in the two sites (Table 6).

The t_{mrt} profile for Site A shows that measurement points near large water bodies actually exhibited the highest t_{mrt} values. The corresponding SVF values are also relatively higher, when compared to the other points. Points 1 and 2 for Site B, while located in the park, show higher t_{mrt} profiles when compared to other measurement points within the park.

Measurement points in Site A show significantly higher readings than those in Site B. This may be due to the fact that there are much more trees in Site B than in Site A, and that the primary function of the trees is to block direct short wave radiation from the sun, rather than reduce the temperature by means of evapotranspiration. The corresponding SVF also supports the correlation between SVF value and t_{mrt}.

3.5. Estimation of t_{mrt} in the absence of sunlight

In the absence of sunlight, the t_{mrt} values do not fluctuate greatly. This is evident in the diurnal profiles of the two Sites (Fig. 8). The difference between the maximum and minimum t_{mrt} values for Site A in the absence of sunlight is 1.3 °C. The difference between the maximum and minimum t_{mrt} values for Site B in the absence of sunlight is 1.0 °C.

The t_{mrt} profile is plotted at 5 min intervals from 00:00 UTC to 07:00 UTC for both sites (Fig. 10). The t_{mrt} profile is fairly constant for Site A, with a slight dip from 05:15 UTC onwards. This dip is observed until 05:35 UTC, and the average decrease in t_{mrt} is 1.6 °C. The dip is more frequent and significant for Site B. The first dip occurred from 01:25 UTC to 01:55 UTC. Point 5 exhibited the largest drop of 4.8 °C. The second dip occurred from 03:50 UTC to 04:15 UTC. Point 5 exhibited the largest drop of 2.2 °C. The final dip occurred from 04:55 UTC to 05:10 UTC. Point 5 exhibited the largest drop of 2.4 °C. Site B was surveyed to understand the drop in t_{mrt} from 01:25 UTC to 01:55 UTC. Wind and air temperature data was observed to have remained constant during the said period. Cloud cover was minimal. Infrared red imaging was used to survey the measurement site for changes in surface temperature. Measurements of the leave surface of trees indicate a slight decrease in temperature of less than 2.0 °C (Fig. 11).

4. Conclusion

4.1. Recalibration of globe thermometer for use in the tropical urban environment

The first objective of this study is to assess the feasibility of deploying 40 mm globe thermometers outdoors for t_{mrt} measurement in the tropics. The 40 mm ping pong ball is preferred over the original 38 mm due to the decrease in availability of 38 mm ping pong balls. Two methods are used to collect t_{mrt} measurements. Method A involves radiant flux measurements via a net radiometer, Method B involves using the 40 mm grey globe thermometer with a mean convection coefficient as stated in ISO 7726:1998 [14]. 5 min mean values are used to compare readings from the two methods. The advantage of using Method A is that long wave and short wave radiation can be measured and analysed in different directions. Method B, while unable to measure long wave and short wave radiation, can provide estimates of t_{mrt} if air temperature and wind velocity data are provided.

Initial comparisons indicate that the t_{mrt} values obtained via the customised globe thermometers and ISO 7726:1998 [14] deviated acutely from measurements obtained via the net radiometer. By redefining the mean convection coefficient of the 40 mm globe thermometer, a recalibrated Equation (4) for the 40 mm grey globe thermometer is obtained. The difference between Method A and Method B with the recalibrated formula is generally small. By conducting the measurement at two different sites, results show that Equation (4) is valid for typical outdoor conditions in Singapore. The air velocity ranges between 0 m s^{-1} and 4 m s^{-1}, and the incoming solar radiation of up to 1300 W m^{-2}. Remaining errors may be attributed to instrumentation errors from the net radiometer and the 40 mm grey globe thermometers (e.g. location of sensors, etc.). The albedo of the 40 mm grey globe may have caused the slight underestimation of short-wave radiation, and it is recommended that a darker shade of grey (lower albedo) be tested for improved results.

In this study, a net radiometer is used for the calibration of the customised globe thermometer. One area of concern is the level uncertainty in measuring t_{mrt} with this method due to the

fluctuating nature of radiation fluxes as well as the sensitivity of the net radiometer. A previous study conducted by d'Ambrosio Alfano et al. (2013) has shown that measurement results from different net radiometers may vary significantly due to equipment specifications alone [39]. This variation, while in compliance to ISO standards, will affect the subsequent results of thermal comfort indices when used as input variables [22,40]. Dell'Isola et al. (2012) concluded from a study of measurement uncertainties on thermal environment assessment that the globe thermometer and net radiometer only produced similar results for the Predicted Mean Vote (PMV) only in conditions of low radiation asymmetry [41]. Since the customised globe thermometer for this study is meant for outdoor deployment and will be subjected to high levels of short wave radiation, further validation using different t_{mrt} measurement methods is recommended.

This study has validated the application of the 40 mm grey globe thermometer in the tropical urban environment. Net radiometers can be used for accurate measurements of t_{mrt} but the equipment is bulky and requires a constant AC power source. The availability and portability of the customised globe thermometer makes it a convenient tool for outdoor t_{mrt} measurements. Since the values of air velocity, globe and air temperature are needed for Equation (4), usage of the 40 mm grey globe thermometers on a large scale for extended periods would entail the concurrent deployment of anemometers and air temperature sensors.

The revised mean convection coefficient provides the possibility of outdoor deployment of the 40 mm globe thermometer together with a data-logger for an extended period of time at an urban scale. Moreover, the availability and portability, coupled with relative ease of deployment makes it an ideal tool for outdoor field measurements.

4.2. Characteristics of t_{mrt} profile in the urban environment

The second objective of this study is to observe t_{mrt} fluctuations in the urban environment in relation to urban constituents and to determine reasonable expectations for any attempts at reduction. Fig. 8 shows that the t_{mrt} profile does not differ significantly in the absence of sunlight. Therefore, any attempt at lowering the t_{mrt} should logically be done only in daytime.

Since the reduction of t_{mrt} is subjected to environmental limitations, it is important to identify a reasonable extent of influence to any proposed intervention via passive methods. Fig. 8 shows that the peak t_{mrt} at Site A for Point 4 has the potential to be lowered, and by drawing reference to Point 9, a reasonable value would be by around 25 °C to approximately 43 °C. A possible way to quantify the reduction is by comparison of the SVF values of both points (Fig. 12), and to propose additions to the landscape to lower the SVF (e.g. planting more trees).

Strategic placement of vegetation is essential for achieving the desired amount of t_{mrt} attenuation. Fig. 8 shows that although the type and quantity of trees in Site B are similar throughout the park, the diurnal t_{mrt} profiles can be different. Measurement Point 1 exhibits a higher maximum t_{mrt} value, as well as at a much later time than Measurement Point 3. This may be due to the fact that Measurement Point 1 is next to a large clearing (water body) and is exposed to direct sunlight in the late afternoon. As each measurement point is isolated and analysed, it is possible to critique the t_{mrt} attenuation potential of the surrounding vegetation.

The dip in t_{mrt} profile in the absence of sunlight for Site B may be due to the large amount of vegetation found in Site B. There is a significant amount of vegetation in the park, and the trees may be the cause of the dip. While a slight decrease in surface temperature of the leaves are observed, it may not fully account for the dip. A more detailed study is required to ascertain the cause of the dips.

Most estimations of t_{mrt} are made in the form of spot readings and do not cover large areas. This presents a challenge to the comparison of t_{mrt} in different urban typologies. Diurnal measurements of t_{mrt} can help urban planners to appreciate the quality of outdoor urban spaces in view of thermal comfort. With the customised globe thermometer, air temperature sensor and anemometer, large scale measurements of the urban outdoor environment can be made. Measurement points with high values of t_{mrt} can be identified and lowered if necessary, so as to improve the quality of outdoor urban space.

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